



Rail Wear Trending & Analysis

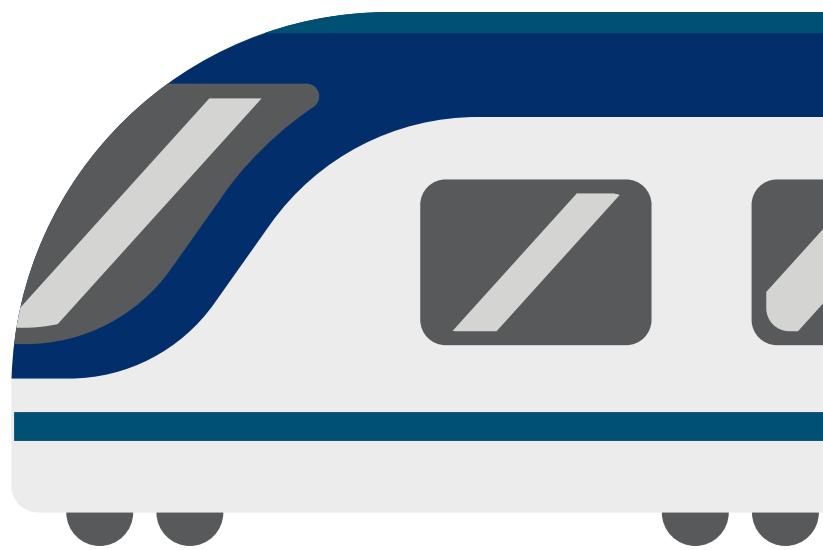
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SBS Transit is one of Singapore's two public transport operators, playing a key role in our nation's daily commute.

It operates the North East Line (NEL), the Downtown Line (DTL), as well as the Sengkang-Punggol LRT, while also being Singapore's largest public bus operator.

Methodology



Cleaned and integrated rail wear, replacement history, and curvature data from Excel data given

Structured data by time since last replacement, track zones, and curve radius

Built regression models for wear prediction across track types (straight, standard curve, premium curve)

Developed RailWatch dashboard to visualise data and enable continuous forecasting with new inputs

*All illustrations were made using sample data

DT 1

The Problem

Rail wear data across SBS Transit's NEL and DTL lines is collected but not systematically analysed for predictive maintenance. Without a forecasting tool, **SBS remains reactive to wear.**

This complicates maintenance planning, particularly when accounting for manpower and scheduling for track access.

NEL & DT

Average Daily Ridership of **1.2M**

DT 2

Rail Wear Over Time

SAMPLE DATA

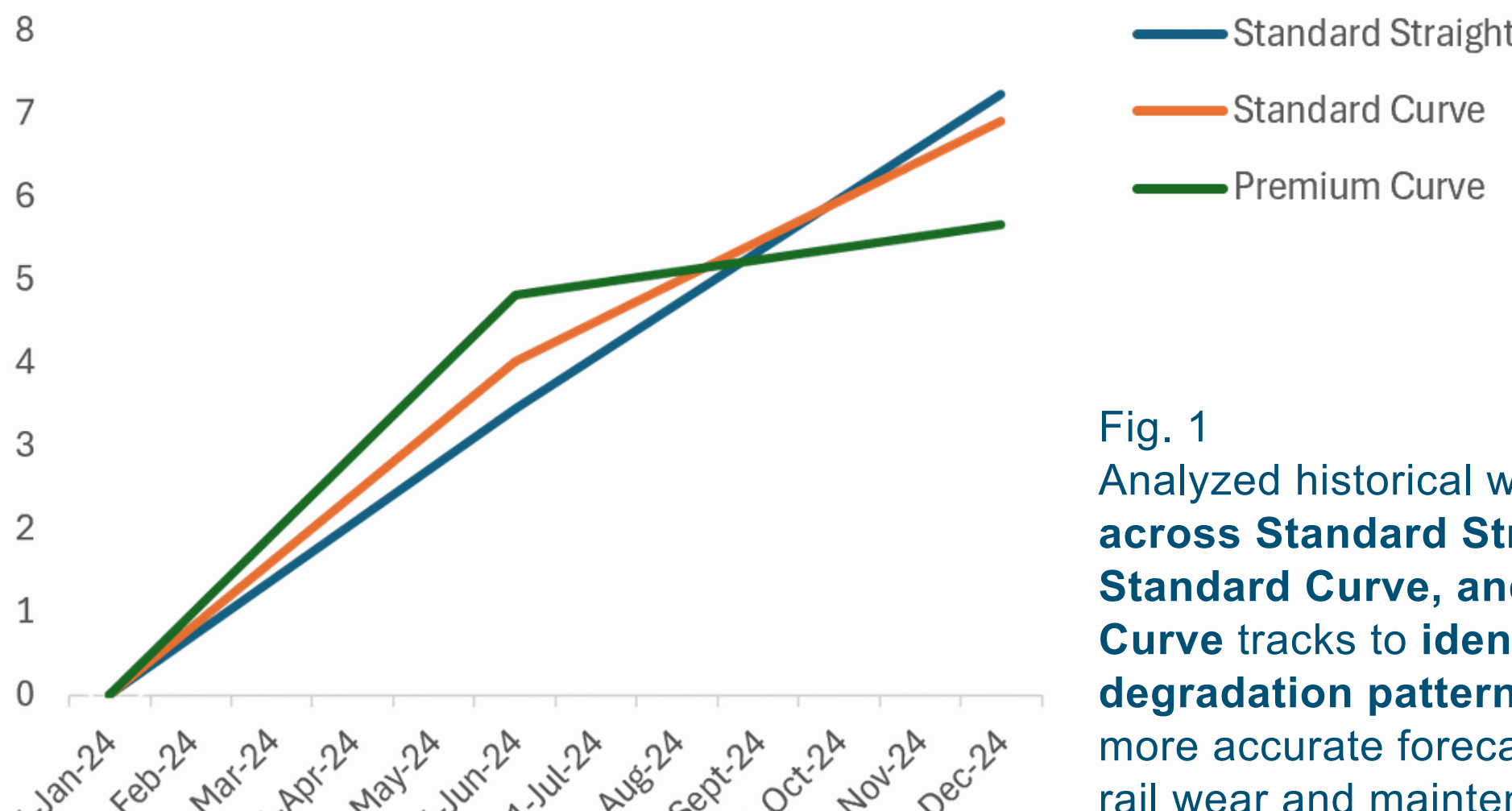


Fig. 1 Analyzed historical wear trends across **Standard Straight**, **Standard Curve**, and **Premium Curve** tracks to identify distinct degradation patterns, enabling more accurate forecasting of future rail wear and maintenance needs.

DT 3

Wear Zone Classification

SAMPLE DATA

< 3 mm	Acceptable
3–5 mm	Monitor
5–6 mm	Plan Repair
6–7 mm	Urgent
≥ 7 mm	Critical

Key Terms

- Chainage:** Specific location along the rail used as a reference point in the forecasting model
- Standard Straight:** Straight track sections using standard rail material
- Standard Curve:** Gentler curves using standard rail material
- Premium Curve:** Sharp curves using harder, premium rail material

DT 4

Rail Health

SAMPLE DATA

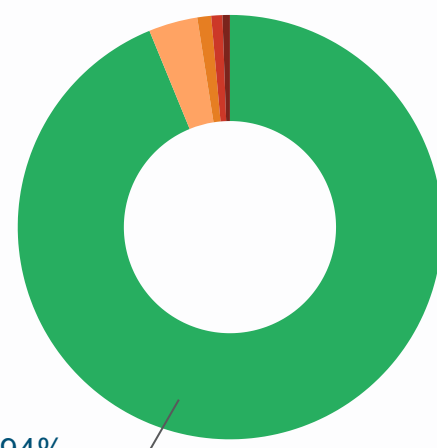


Fig. 2 A large majority (94%) fall within acceptable limits, indicating **overall good system health.**

NE 1

Final Model

Current System

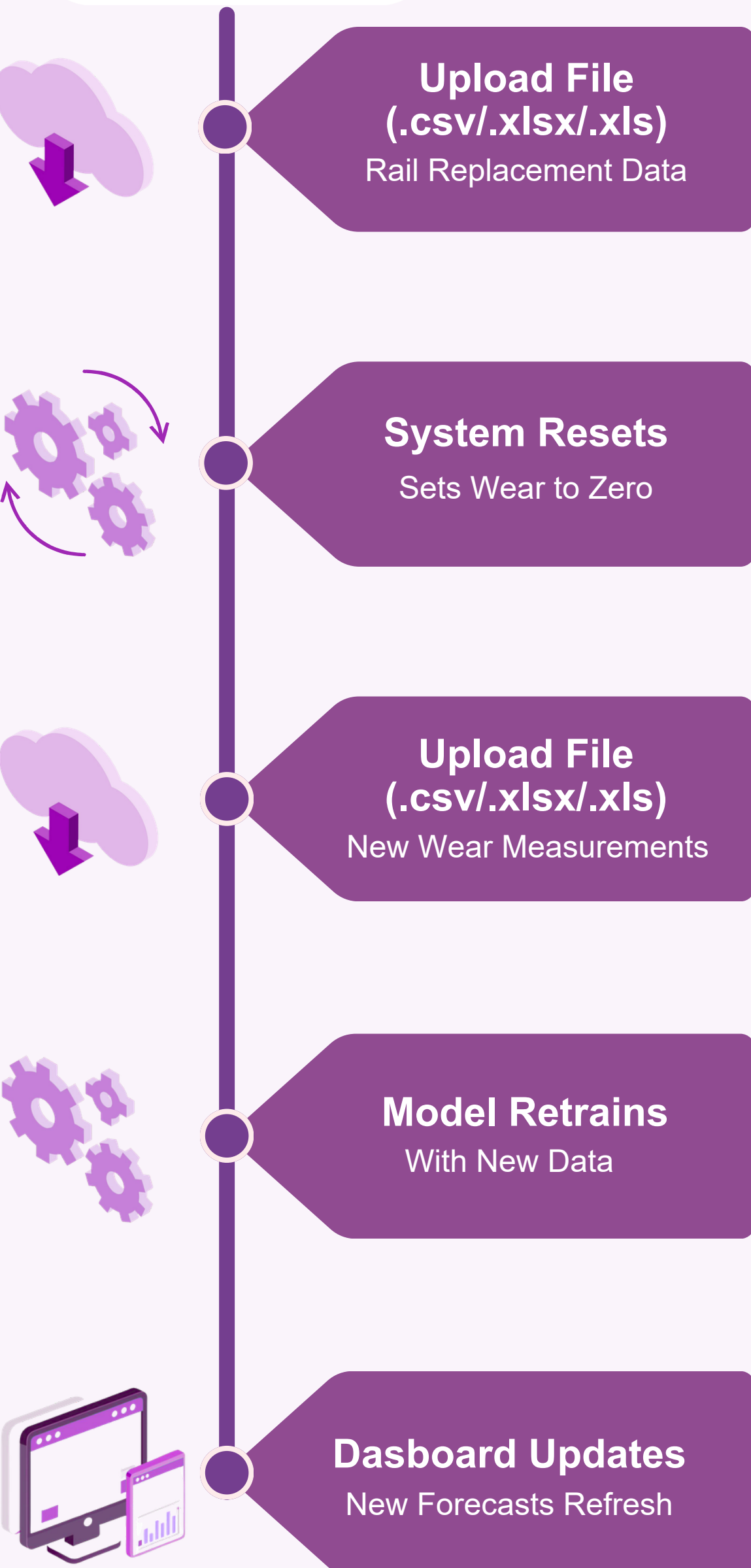
Corrective Maintenance



Our Goal

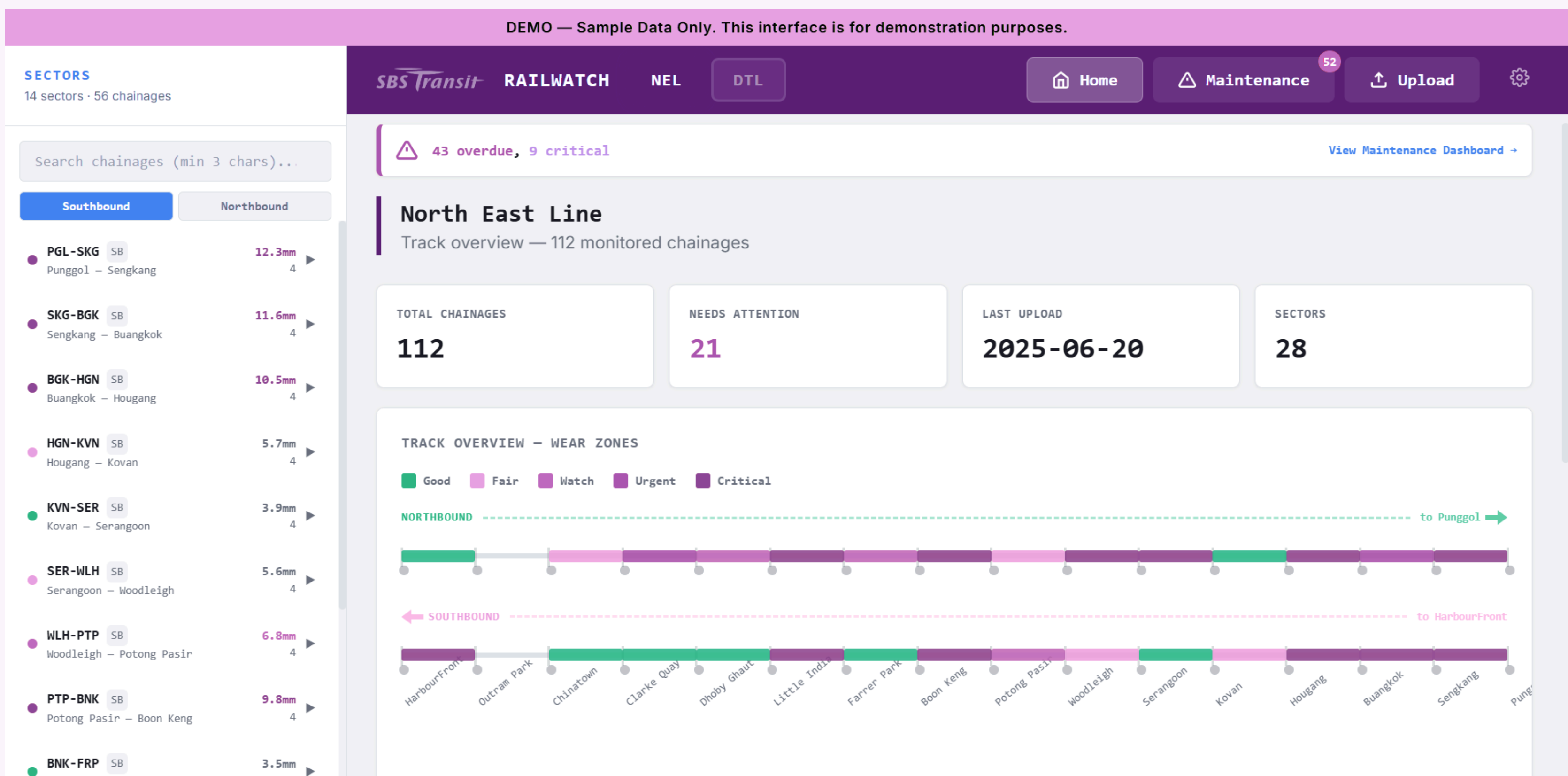
Predictive Maintenance

Our System



NE 2

RAILWATCH



On curved tracks, inner rails wear more at the crown (0°), while outer rails wear more on the side (90°). RailWatch classifies this automatically using curve direction.

Recency Weighting (recent inspections influence the model more):

0–6 months: 1.0× 6–12 months: 0.8× 12–24 months: 0.5× >24 months: 0.3×

Track Coverage

Line	Tracks	Angular Positions Measured
NEL (North East Line)	To and From Harbourfront	0°, 90°
DTL (Downtown Line)	To and From Expo	0°, 22.5°, 45°, 67.5°, 90°

Model Details

- Each position predicted independently; earliest to hit 5* mm sets the chainage repair date
- Falls back to simple linear regression if curvature term produces an unstable slope
- Model automatically retrains on every new inspection upload, improves continuously

Final Model Considerations

Global Pooling Advantage

Pooling thousands of similar chainages gives each model far more training samples than per-chainage fits, producing stable wear-rate estimates even for sections with sparse history.

Curve Rail Differentiation

Curved sections model inner and outer rails separately to reflect directional wear patterns: the inner rail shows greater wear at the crown (0°), while the outer rail exhibits higher lateral wear(90°).

Automated Wear Reset

When new replacement data is uploaded, replaced sections of rail are identified by their chainages and their wear values are reset to 0.

Weighted by Recency

Sample weights biased toward recent inspections keep the model responsive to current wear conditions without discarding historical data, a balance between stability and adaptability.

“Intelligent Dashboard for Predictive Rail Wear”

The Prediction Machine

RailWatch uses 25 global regression models trained by pooling data across all chainages of the same track category. Three formula variants handle straight track and two types of curved track, curved sections include a curvature term (tR) to account for the increased wear on the outer rail due to centripetal force.

Cat.	Track Type	Regression Formula
1	Standard Straight	$\text{wear}(t) = \beta_0 + \beta_1 \cdot t$
2	Standard Curve	$\text{wear}(t) = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot (t/R)$
3	Premium Curve	$\text{wear}(t) = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot (t/R)$

β_0 = baseline wear (mm) · β_1 = linear wear rate (mm/yr) · β_2 = curvature term · t = years · R = curve radius (m)

Assumptions

- Wear progresses smoothly over time
- Historical patterns persist into the future
- Measurement process is consistent
- Track geometry remains unchanged
- Operating conditions are stable

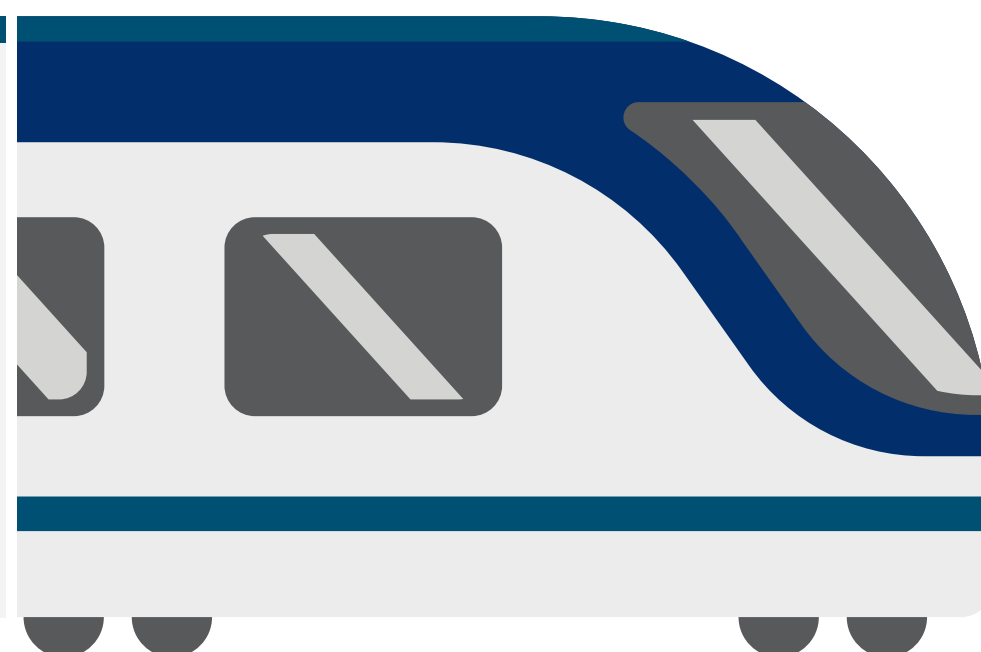
Limitations

- Limited history for DTL (only one inspection date per section)
- Replacements not purely wear-driven
- Possible data quality issues

Conclusion

Analysed rail wear across standard straight, standard curve, and premium curve tracks to identify distinct degradation patterns.

Developed an intuitive dashboard that highlights sections requiring attention, incorporates company-specific requirements, and provides data-driven forecasts to support predictive maintenance planning.



Scan to Try Our Interface

